Milestone-03: Geographical Distribution Analysis

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1. Project Description

Milestone-3 focuses on analyzing the geographical distribution of natural disaster declarations across the United States. The cleaned dataset from Milestone-1 is used to perform geospatial analysis and exploratory data analysis (EDA). Disaster records are aggregated by state and incident type to identify regions with higher disaster frequency. Various visualizations such as choropleth maps, bar charts, and stacked charts are created to highlight disaster hotspots and regional patterns. These visualizations help understand how disasters vary across different states and regions of the country.

2. Dataset Used

The dataset used for this milestone is **usnd_cleaned.csv**, which was generated during **Milestone-1** after cleaning the original disaster dataset. It contains records of natural disaster declarations across different states in the United States. The dataset includes key attributes such as **state, incident type, year, month, and declaration date**, which are important for performing geographic analysis. Before analysis, the dataset was validated to ensure consistent state codes and to remove any missing or invalid values. This cleaned dataset enables accurate geographic aggregation and visualization of disaster patterns.

3. Environment Setup

The project was implemented using **Python** in a development environment such as **Jupyter Notebook or VS Code**. Several libraries were used to perform data processing, analysis, and visualization. Key libraries include **Pandas and NumPy** for data manipulation, **Matplotlib and Seaborn** for creating charts, and **Plotly** for generating interactive geographic visualizations such as choropleth maps. For geospatial analysis, libraries like **GeoPandas and Folium** can also be used to create interactive maps. These tools provide an efficient environment for performing geographic analysis and visualizing disaster patterns across different states.

4. Data Exploration

Before performing geographic analysis, the dataset was explored to understand its structure, verify data quality, and identify important attributes required for the analysis. Data exploration helps ensure that the dataset is properly loaded and ready for further processing.

4.1 Loading the Dataset

The cleaned dataset **usnd_cleaned.csv** was loaded into the analysis environment using Python. Loading the dataset allows the data to be inspected and prepared for further analysis.

4.2 Inspecting Dataset Structure

The structure of the dataset was examined to understand the number of records, column names, and data types. This step ensures that the dataset contains the required attributes such as **state, incident type, year, and month**.

4.3 Checking State Distribution

The dataset was analyzed to determine the number of unique states present in the disaster records. This helps verify that the dataset contains disaster declarations from multiple states across the United States.

4.4 Examining Disaster Types

The different types of disasters recorded in the dataset were examined. Identifying disaster categories such as floods, hurricanes, storms, and fires helps understand the variety of disasters included in the dataset.

4.5 Identifying Missing Values

The dataset was checked for missing or invalid values, especially in the **state column**, since geographic analysis depends on accurate location information. Any missing values were handled to ensure reliable analysis.

5. Data Preprocessing

Data preprocessing is an essential stage before performing geographic analysis. It ensures that the dataset is clean, structured, and ready for aggregation and visualization. During this stage, the disaster dataset was prepared by verifying data quality, converting necessary fields, and removing inconsistencies.

5.1 Dataset Loading

The cleaned dataset **usnd_cleaned.csv** was loaded into the analysis environment using Python libraries such as Pandas.

Purpose

- To access the dataset for analysis.
- To verify that the data is correctly loaded into the system.

Steps

1. Import the required Python libraries.
2. Load the dataset using Pandas.
3. Display the first few rows of the dataset to confirm successful loading.

5.2 Date Conversion

The declaration date column was converted into datetime format to extract important time-related attributes.

Purpose

- **To extract useful features such as year and month from the declaration date.**
- **To organize disaster records based on time.**

Steps

1. **Convert the declaration date column into datetime format.**
2. **Extract the year from the date column.**
3. **Extract the month from the date column.**
4. **Add the new columns to the dataset.**

5.3 Handling Missing Values

The dataset was checked for missing or invalid values, particularly in the state column, which is required for geographic analysis.

Purpose

- **To ensure that all records contain valid state information.**
- **To prevent errors during geographic aggregation and visualization.**

Steps

1. **Check the dataset for missing values in the state column.**
2. **Identify records with missing or invalid state values.**
3. **Remove or correct these records to maintain data consistency.**

5.4 Feature Verification

Important dataset features were verified to ensure that the required columns were present.

Purpose

- To confirm the availability of key attributes required for geographic analysis.
- To ensure the dataset includes **state, incident type, year, and month** columns.

Steps

1. Display the list of column names in the dataset.
2. Verify that required features exist.
3. Ensure that the column names are consistent and correctly formatted.

5.5 Data Preparation for Geographic Analysis

After cleaning and verifying the dataset, the data was prepared for geographic aggregation and visualization.

Purpose

- To organize the dataset for spatial analysis.
- To ensure the dataset is ready for mapping and disaster pattern identification.

Steps

1. Validate that the dataset contains records from multiple states.
2. Ensure that all required columns are present.
3. Prepare the dataset for grouping and visualization tasks.

6. Proposed System

The proposed system focuses on analyzing and visualizing the geographical distribution of natural disaster declarations across the United States. The system processes the cleaned disaster dataset and applies geographic aggregation and visualization techniques to identify disaster-prone states and regional patterns.

The system workflow involves loading the cleaned dataset, preparing the data for geographic analysis, aggregating disaster declarations by location, and generating visualizations such as maps and charts. These visualizations help reveal spatial patterns and disaster hotspots across different states.

6.1 Input Dataset

The system uses the cleaned dataset usnd_cleaned.csv, which contains records of natural disaster declarations across various states in the United States. This dataset includes attributes such as state, incident type, year, month, and declaration date, which are required for geographic analysis.

6.2 Data Preparation

Before performing geographic analysis, the dataset is prepared by verifying column consistency, converting date values, and removing missing or invalid records. This ensures that the dataset contains accurate and reliable information for analysis.

6.3 Geographic Aggregation

The disaster records are grouped by state to calculate the total number of disaster declarations in each state. This aggregation helps identify states with the highest disaster frequency. Additionally, the dataset is grouped by state and incident type to determine which types of disasters are dominant in specific states.

6.4 Geographic Visualization

After aggregation, the system generates visualizations to represent disaster patterns across states. These visualizations include:

- Choropleth maps to display disaster frequency across states using color intensity.
- Bar charts to compare disaster counts among the top states.
- Stacked bar charts to show the distribution of disaster types within each state.

These visualizations help identify disaster hotspots and highlight regional differences in disaster occurrences.

6.5 Insight Generation

The final stage of the proposed system involves interpreting the visualizations to generate meaningful insights. By analyzing geographic patterns, it becomes possible to identify disaster-prone states, understand regional disaster trends, and recognize dominant disaster types in different areas.

These insights can support better disaster preparedness planning and improve resource allocation for disaster management.

7. Training Pipeline

Features

The training pipeline includes several important features that ensure efficient analysis and visualization of disaster data.

- **Structured Data Processing** – The dataset is processed in a step-by-step pipeline, starting from loading and validation to aggregation and visualization.

- **Geographic Aggregation** – Disaster declarations are grouped by state and incident type to identify disaster-prone areas and dominant disaster types.

- **Visualization Generation** – The pipeline produces multiple visualizations such as choropleth maps, bar charts, and stacked bar charts to represent disaster distribution.

- **Hotspot Identification** – The system highlights regions with higher disaster frequency using geographic maps.

- **Insight Generation** – The visualizations generated through the pipeline help identify spatial patterns and regional disaster trends.

This structured pipeline ensures that the disaster dataset is processed efficiently and that meaningful geographic insights can be extracted from the analysis.

8. Conclusion

In this milestone, the geographical distribution of natural disaster declarations across the United States was analyzed using the cleaned dataset. The dataset was prepared and validated to ensure accurate geographic analysis. Disaster records were aggregated by state and incident type to understand spatial patterns and identify disaster-prone regions.

Various visualizations such as choropleth maps, bar charts, and stacked bar charts were created to represent disaster frequency across different states. These visualizations helped identify disaster hotspots and understand how different disaster types are distributed geographically.

The analysis provides valuable insights into regional disaster trends and highlights states that experience higher disaster occurrences. These insights can support better disaster preparedness, planning, and resource allocation for effective disaster management.